

74HCxx / 74LSxx IC Kit

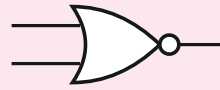
All 22 device types in the kit, drawn as the logic they contain. The bold number on each card is the page — printed in the corner of that page — carrying its full pinout, internal diagram and truth table. ■ 74LS · Schottky TTL ■ 74HC · CMOS



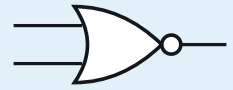
74LS00
Quad 2-input NAND
DIP-14 · [page 3](#)



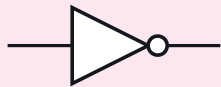
74HC00
Quad 2-input NAND
DIP-14 · [page 4](#)



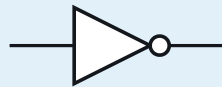
74LS02
Quad 2-input NOR
DIP-14 · [page 5](#)



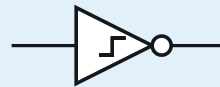
74HC02
Quad 2-input NOR
DIP-14 · [page 6](#)



74LS04
Hex inverter
DIP-14 · [page 7](#)



74HC04
Hex inverter
DIP-14 · [page 8](#)



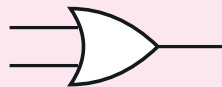
74HC14
Hex Schmitt inverter
DIP-14 · [page 9](#)



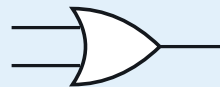
74LS08
Quad 2-input AND
DIP-14 · [page 10](#)



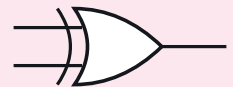
74HC08
Quad 2-input AND
DIP-14 · [page 11](#)



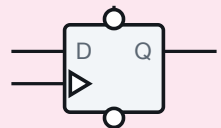
74LS32
Quad 2-input OR
DIP-14 · [page 12](#)



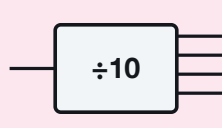
74HC32
Quad 2-input OR
DIP-14 · [page 13](#)



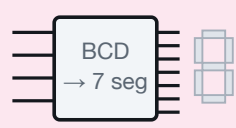
74LS86
Quad 2-input XOR
DIP-14 · [page 14](#)



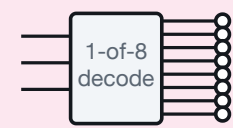
74LS74
Dual D flip-flop
DIP-14 · [page 15](#)



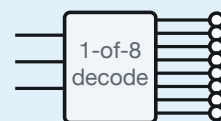
74LS90
Decade counter
DIP-14 · [page 16](#)



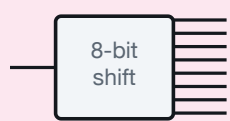
74LS47
BCD to 7-segment
DIP-16 · [page 17](#)



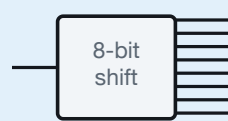
74LS138
3-to-8 decoder
DIP-16 · [page 18](#)



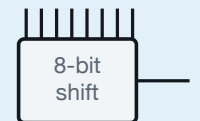
74HC138
3-to-8 decoder
DIP-16 · [page 19](#)



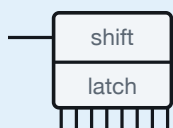
74LS164
Shift register, serial in
DIP-14 · [page 20](#)



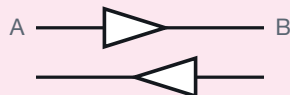
74HC164
Shift register, serial in
DIP-14 · [page 21](#)



74HC165
Shift register, parallel in
DIP-16 · [page 22](#)



74HC595
Shift register + output latch
DIP-16 · [page 23](#)



74LS245
Octal bus transceiver
DIP-20 · [page 24](#)

74HCxx / 74LSxx IC Kit

Pinout and internal function diagram for all 22 device types in the Minidodoca 22-value / 44-piece kit — 2 pieces of each part, plus DIP-14, DIP-16 and DIP-20 sockets. One page per part: the package pinout, the logic that is actually inside the die, and the truth table.

THE TWO FAMILIES IN THIS KIT

74LS – low-power Schottky TTL

- VCC **4.75 – 5.25 V only** (5 V, $\pm 5\%$).
- Input LOW ≤ 0.8 V, input HIGH ≥ 2.0 V.
- A floating input **reads as HIGH** and picks up noise — tie unused inputs to VCC through 1 k Ω , or to GND.
- Output sinks ~ 8 mA but sources only ~ 0.4 mA — wire LEDs from VCC *into* the pin, not from the pin to ground.
- Draws several mA even when nothing is switching.

74HC – high-speed CMOS

- VCC **2 – 6 V**. Runs at 3.3 V or 5 V.
- Switching threshold sits near VCC/2; inputs draw essentially no current.
- Inputs must still never be left open — an open CMOS input oscillates and heats the chip.
- Outputs source *and* sink ± 4 mA (± 6 mA on the '245 and '595), swinging rail to rail.
- Near-zero idle current, but a 74HC input is **not** guaranteed to see a 74LS HIGH of 2.4 V — use a 74HCT part when an LS output has to drive a CMOS input.

HOW TO READ THESE PAGES

Pinout drawings

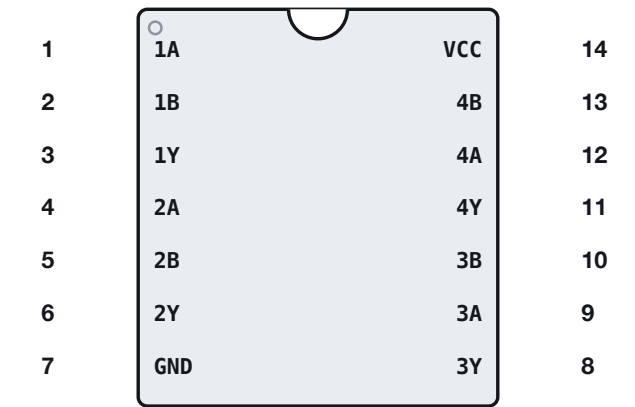
- Drawn **top view**: notch up, pin 1 at the top left, numbering counter-clockwise. That is exactly how the part looks sitting in a breadboard.
- Power pins are not always the corners — the 74LS90 puts VCC on pin 5 and GND on pin 10.
- NC means *no connection*: leave it open, do not use it as a tie point.

Function diagrams

- Numbers in circles are **package pin numbers**.
- A bubble on a gate input or output, or a bar over a name ($\overline{\text{CLR}}$), means that signal is **active LOW**.
- A small triangle at a flip-flop input marks the **clock**; C1 / 1D is the standard IEC notation for "this D input is controlled by clock 1".

Before you power anything up: put a 0.1 μF ceramic capacitor between VCC and GND right at each chip, check the notch orientation, and never let two outputs drive the same node. Series-resistor every LED (220 – 470 Ω at 5 V).

PINOUT — DIP-14

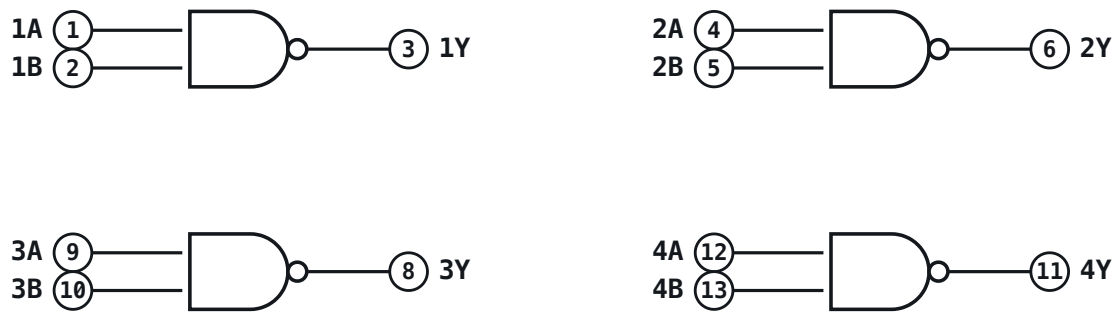


DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

Function	$Y = \overline{A \cdot B}$
Gates per package	4 independent 2-input gates
Power	VCC = pin 14, GND = pin 7

FUNCTION DIAGRAM — WHAT IS INSIDE



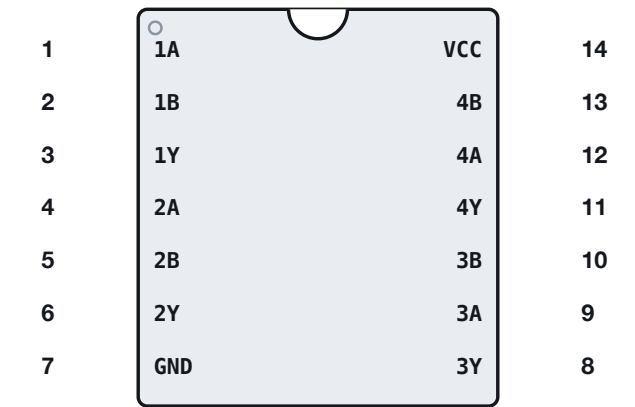
Four independent 2-input NAND gates. VCC = pin 14, GND = pin 7.

LOGIC TABLES

TRUTH TABLE (EACH OF THE FOUR GATES)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

PINOUT — DIP-14

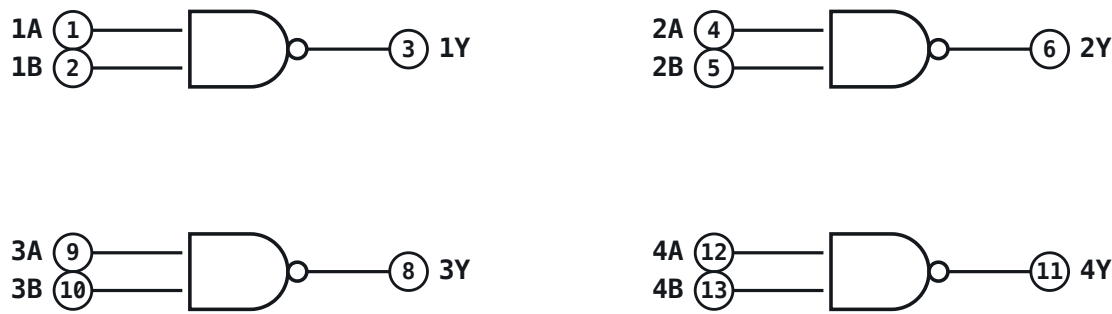


DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

Function	$Y = \overline{A \cdot B}$
Gates per package	4 independent 2-input gates
Power	VCC = pin 14, GND = pin 7

FUNCTION DIAGRAM — WHAT IS INSIDE



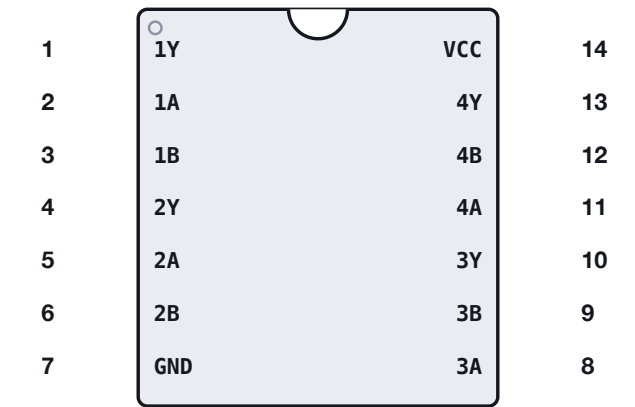
Four independent 2-input NAND gates. VCC = pin 14, GND = pin 7.

LOGIC TABLES

TRUTH TABLE (EACH OF THE FOUR GATES)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

PINOUT — DIP-14

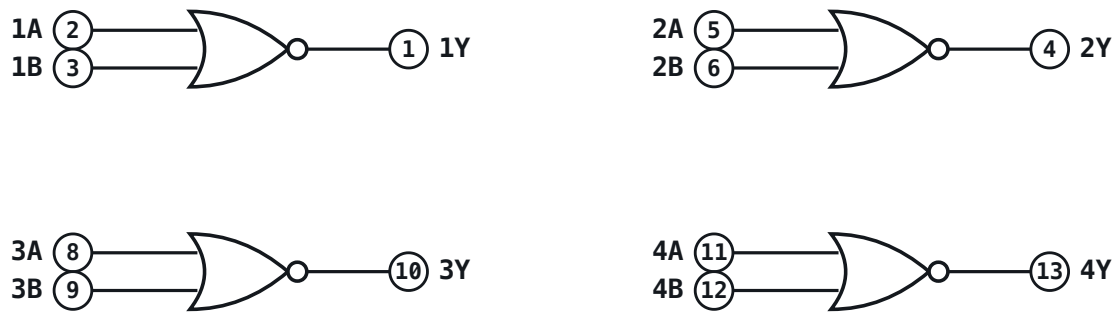


DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

Function	$Y = \overline{A + B}$
Gates per package	4 independent 2-input gates
Power	VCC = pin 14, GND = pin 7

FUNCTION DIAGRAM — WHAT IS INSIDE



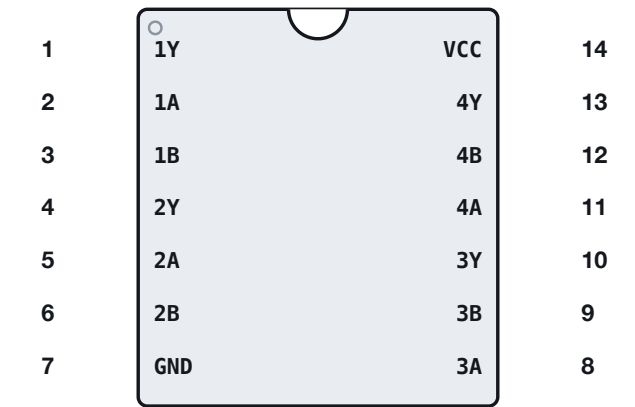
Four independent 2-input NOR gates. VCC = pin 14, GND = pin 7.

LOGIC TABLES

TRUTH TABLE (EACH OF THE FOUR GATES)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

PINOUT — DIP-14

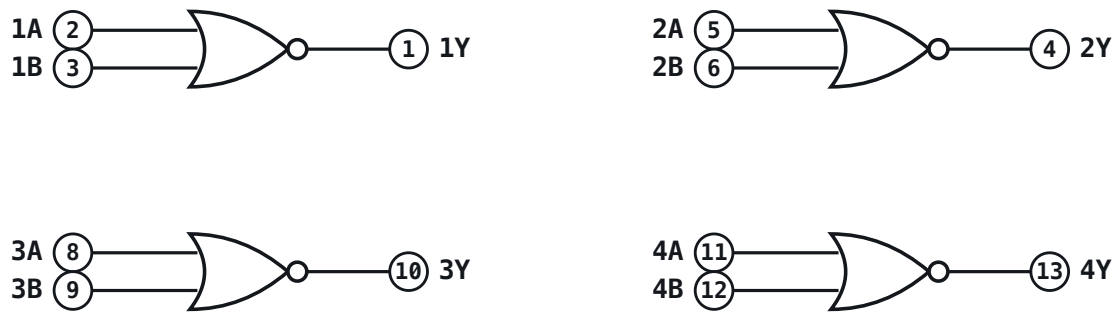


DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

Function	$Y = \overline{A + B}$
Gates per package	4 independent 2-input gates
Power	VCC = pin 14, GND = pin 7

FUNCTION DIAGRAM — WHAT IS INSIDE



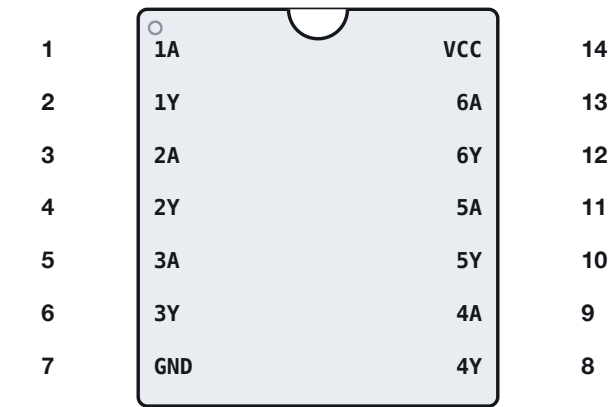
Four independent 2-input NOR gates. VCC = pin 14, GND = pin 7.

LOGIC TABLES

TRUTH TABLE (EACH OF THE FOUR GATES)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

PINOUT — DIP-14

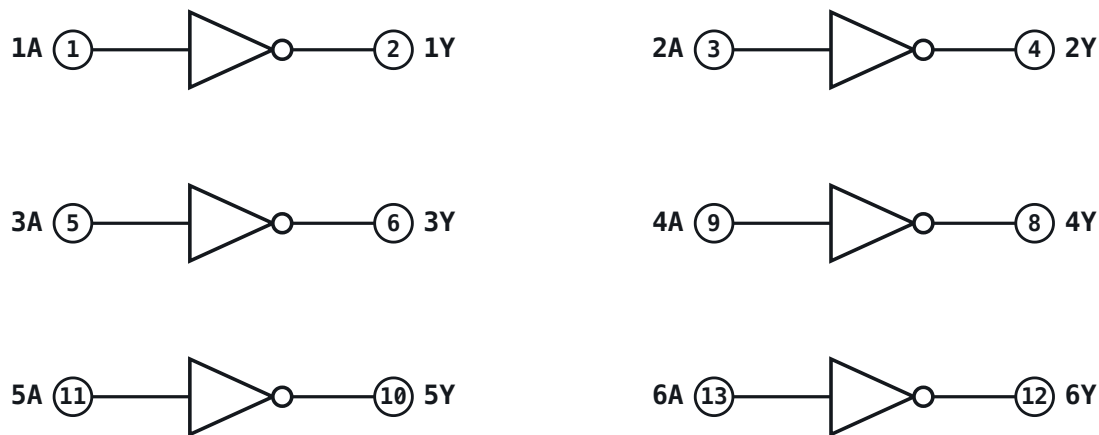


DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

Function	$Y = \overline{A}$
Inverters per package	6
Power	VCC = pin 14, GND = pin 7

FUNCTION DIAGRAM — WHAT IS INSIDE



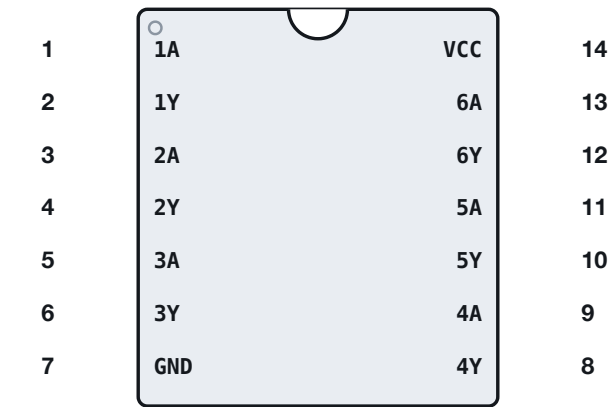
Six independent inverters. VCC = pin 14, GND = pin 7.

LOGIC TABLES

TRUTH TABLE (EACH OF THE SIX INVERTERS)

A	Y
0	1
1	0

PINOUT — DIP-14

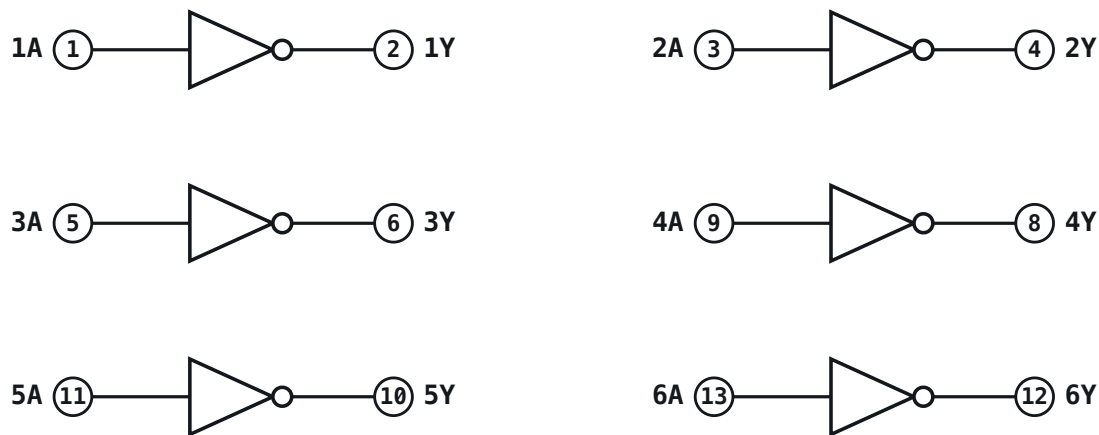


DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

Function	$Y = \overline{A}$
Inverters per package	6
Power	VCC = pin 14, GND = pin 7

FUNCTION DIAGRAM — WHAT IS INSIDE



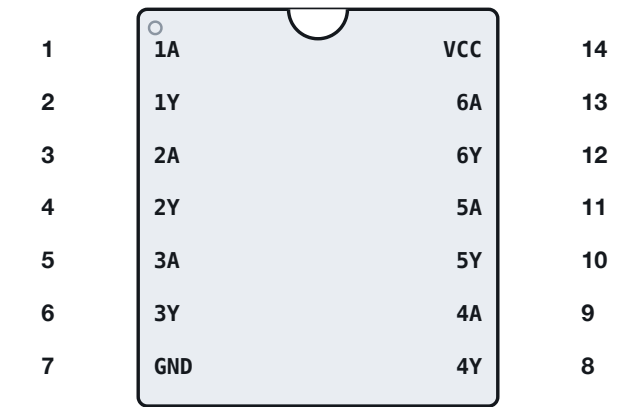
Six independent inverters. VCC = pin 14, GND = pin 7.

LOGIC TABLES

TRUTH TABLE (EACH OF THE SIX INVERTERS)

A	Y
0	1
1	0

PINOUT — DIP-14

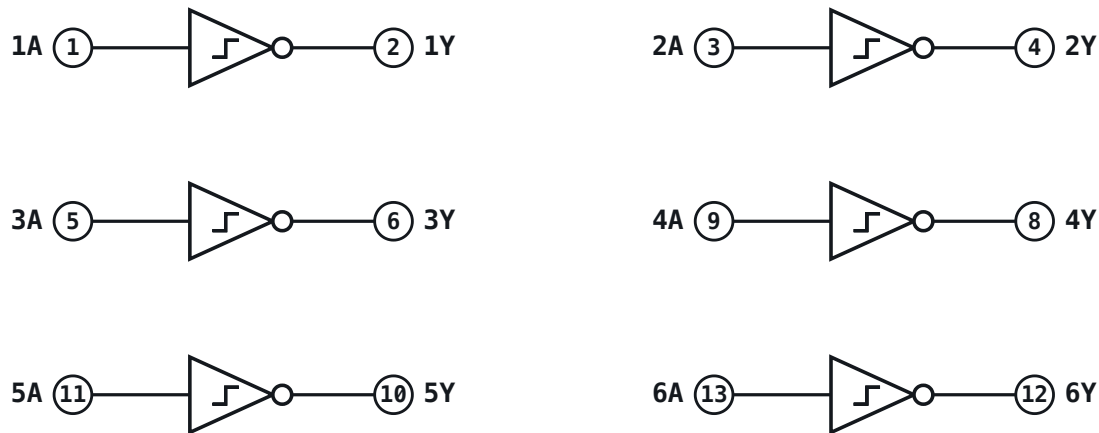


DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

Function	$Y = \overline{A}$
Inverters per package	6
Power	VCC = pin 14, GND = pin 7
Why Schmitt	Cleans up slow, noisy or bouncing edges — the input must cross a different threshold going up than coming down.

FUNCTION DIAGRAM — WHAT IS INSIDE



Six independent inverters with Schmitt-trigger inputs. VCC = pin 14, GND = pin 7.

LOGIC TABLES

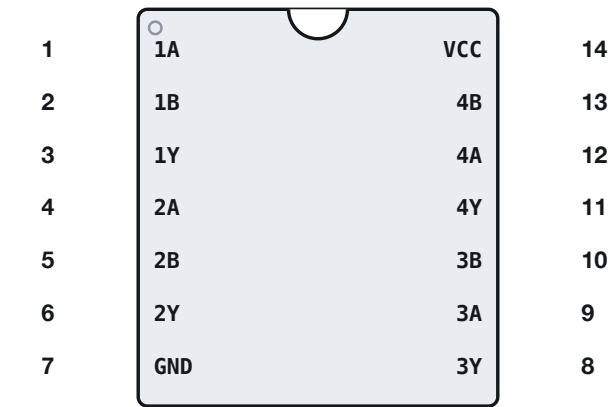
TRUTH TABLE (EACH OF THE SIX INVERTERS)

A	Y
0	1
1	0

SCHMITT-TRIGGER THRESHOLDS (VCC = 4.5 V, 25 °C)

Parameter	Typ
VT+ (rising)	2.7 V
VT- (falling)	1.6 V
Hysteresis	1.1 V

PINOUT — DIP-14

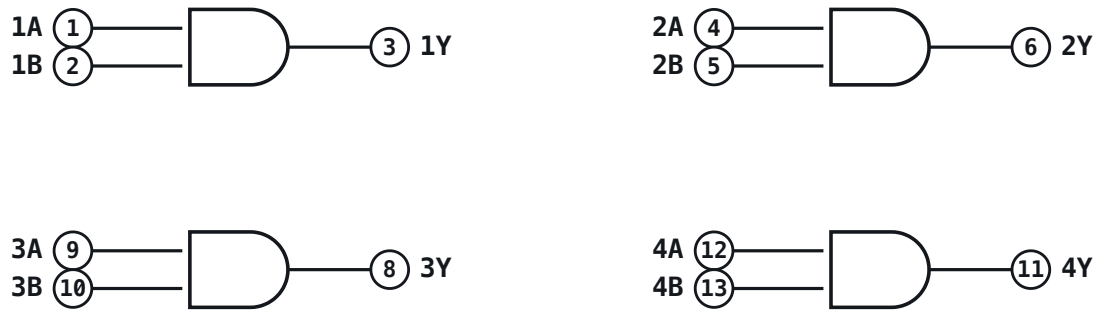


DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

Function	$Y = A \cdot B$
Gates per package	4 independent 2-input gates
Power	VCC = pin 14, GND = pin 7

FUNCTION DIAGRAM — WHAT IS INSIDE



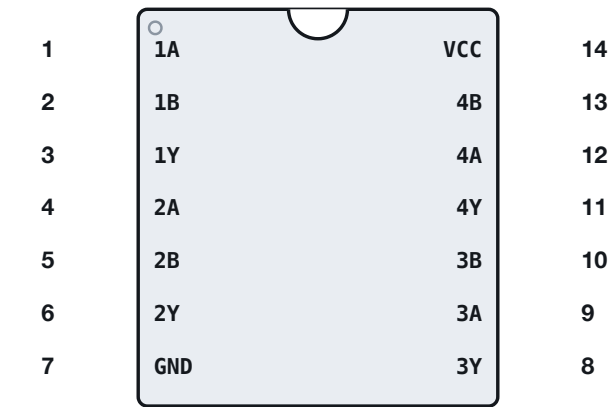
Four independent 2-input AND gates. VCC = pin 14, GND = pin 7.

LOGIC TABLES

TRUTH TABLE (EACH OF THE FOUR GATES)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

PINOUT — DIP-14

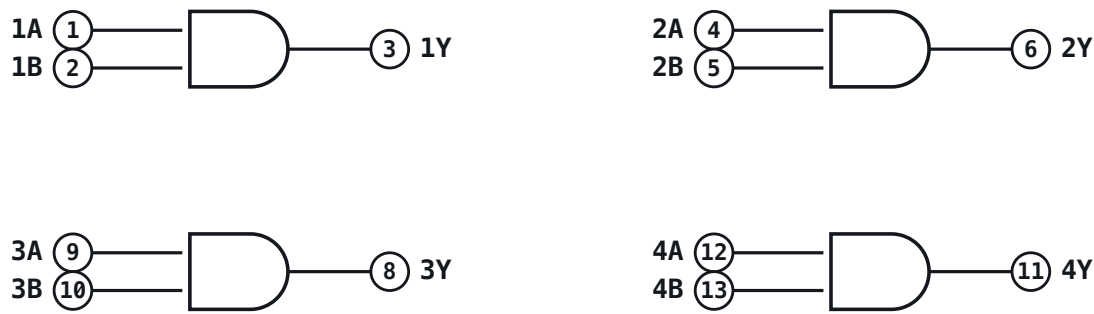


DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

Function	$Y = A \cdot B$
Gates per package	4 independent 2-input gates
Power	VCC = pin 14, GND = pin 7

FUNCTION DIAGRAM — WHAT IS INSIDE



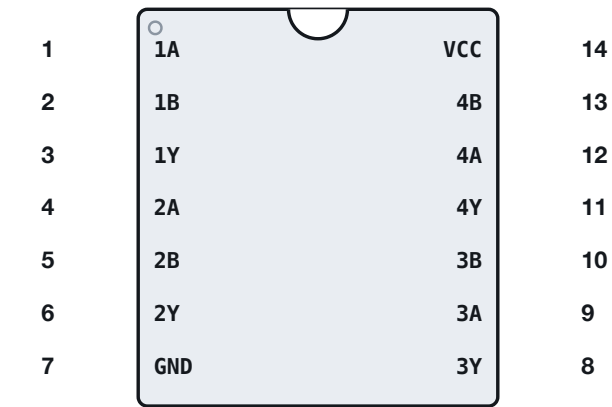
Four independent 2-input AND gates. VCC = pin 14, GND = pin 7.

LOGIC TABLES

TRUTH TABLE (EACH OF THE FOUR GATES)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

PINOUT — DIP-14

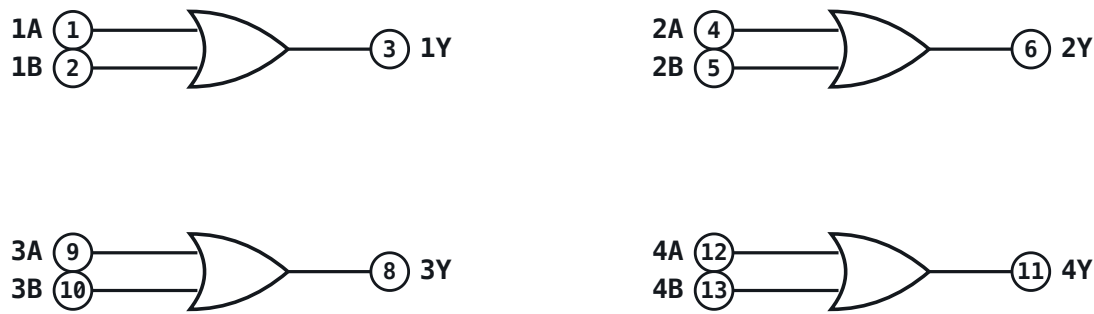


DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

Function	$Y = A + B$
Gates per package	4 independent 2-input gates
Power	VCC = pin 14, GND = pin 7

FUNCTION DIAGRAM — WHAT IS INSIDE



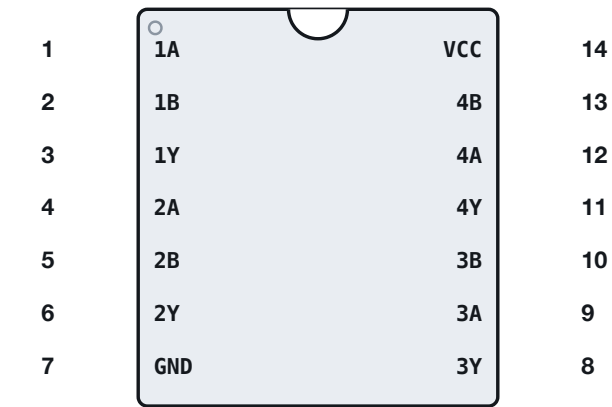
Four independent 2-input OR gates. VCC = pin 14, GND = pin 7.

LOGIC TABLES

TRUTH TABLE (EACH OF THE FOUR GATES)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

PINOUT — DIP-14

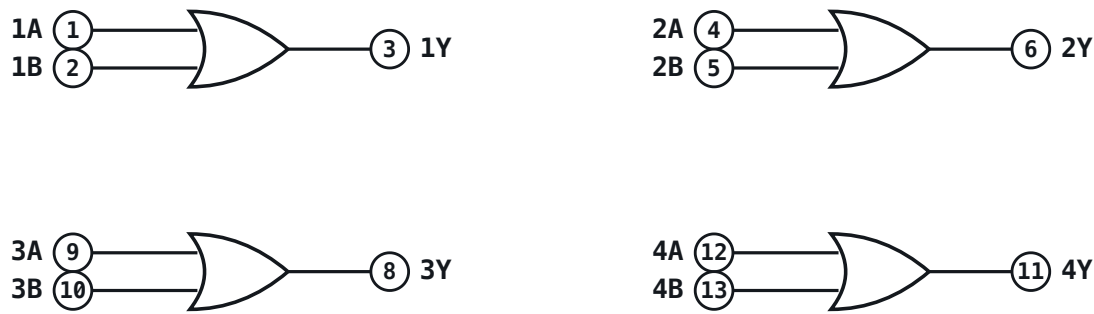


DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

Function	$Y = A + B$
Gates per package	4 independent 2-input gates
Power	VCC = pin 14, GND = pin 7

FUNCTION DIAGRAM — WHAT IS INSIDE



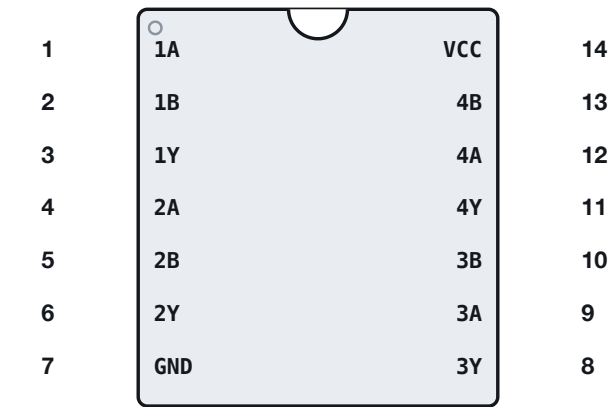
Four independent 2-input OR gates. VCC = pin 14, GND = pin 7.

LOGIC TABLES

TRUTH TABLE (EACH OF THE FOUR GATES)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

PINOUT — DIP-14

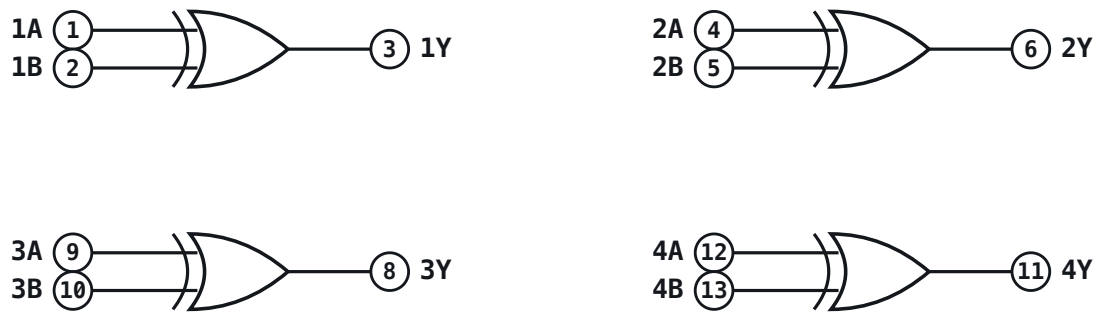


DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

Function	$Y = A \oplus B$
Gates per package	4 independent 2-input gates
Power	VCC = pin 14, GND = pin 7

FUNCTION DIAGRAM — WHAT IS INSIDE



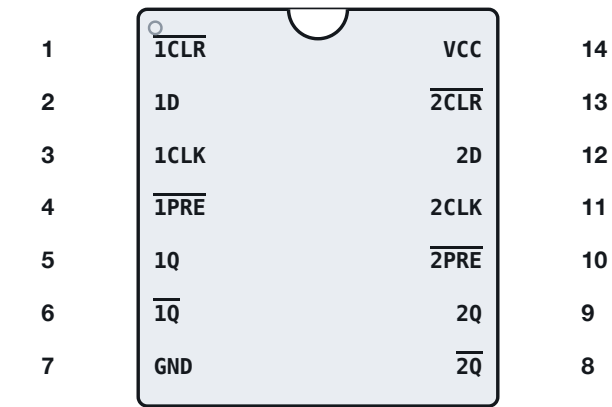
Four independent 2-input XOR gates. VCC = pin 14, GND = pin 7.

LOGIC TABLES

TRUTH TABLE (EACH OF THE FOUR GATES)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

PINOUT — DIP-14



DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

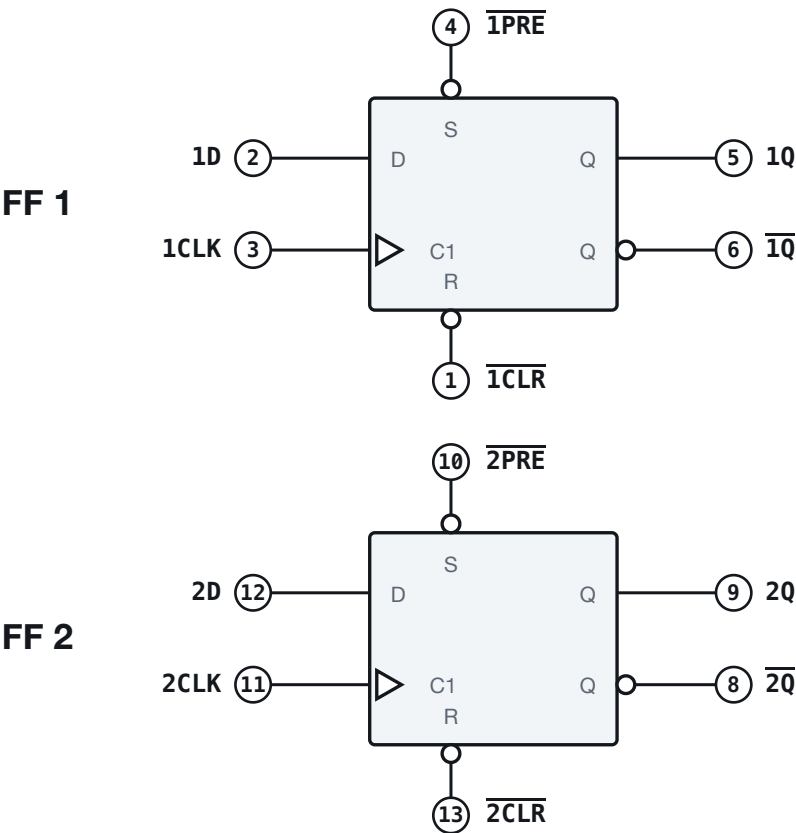
- Trigger** Positive edge (LOW → HIGH) on CLK
- PRE / CLR** Asynchronous, active LOW, override the clock
- Typical use** Divide-by-2, edge synchroniser, 1-bit register
- Power** VCC = pin 14, GND = pin 7

LOGIC TABLES

FUNCTION TABLE					
PRE	CLR	CLK	D	Q	Q̄
L	H	X	X	H	L
H	L	X	X	L	H
L	L	X	X	H*	H*
H	H	↑	H	H	L
H	H	↑	L	L	H
H	H	L	X	hold	hold

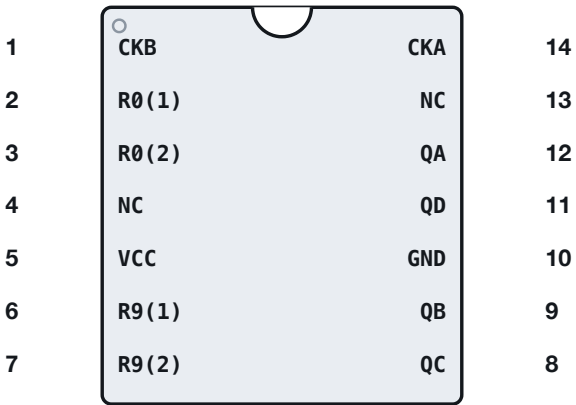
* Both LOW is a forbidden state — the outputs are not complementary.

FUNCTION DIAGRAM — WHAT IS INSIDE



Two independent positive-edge-triggered D flip-flops.
PRE and CLR are asynchronous and active LOW. VCC = pin 14, GND = pin 7.

PINOUT — DIP-14



DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

- Trigger** Negative edge (HIGH → LOW) on CKA / CKB
- BCD decade** Clock into CKA (14), tie QA (12) → CKB (1). Outputs QD QC QB QA.
- Bi-quinary ÷10** Clock into CKB (1), tie QD (11) → CKA (14). Gives a 50 % duty output on QA.
- Power** VCC = pin 5, GND = pin 10 — not the usual corners!

LOGIC TABLES

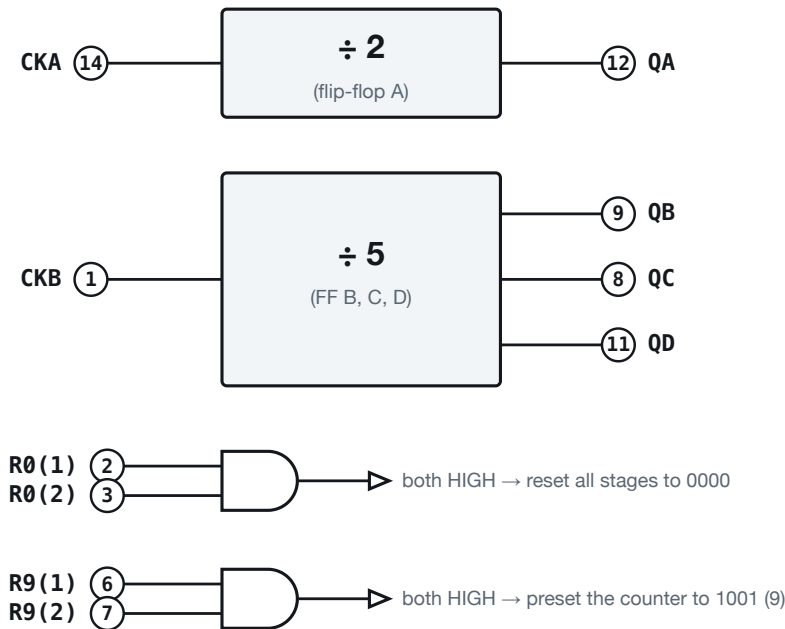
BCD COUNT SEQUENCE (CKA IN, QA → CKB)

	0	1	2	3	4	5	6	7	8	9
QD	0	0	0	0	0	0	0	0	1	1
QC	0	0	0	0	1	1	1	1	0	0
QB	0	0	1	1	0	0	1	1	0	0
QA	0	1	0	1	0	1	0	1	0	1

RESET / SET INPUTS

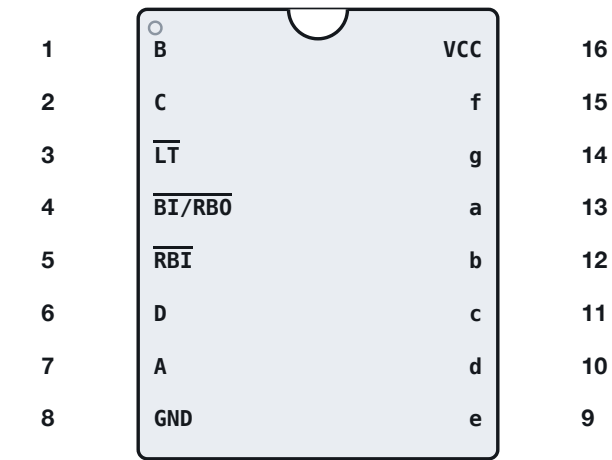
R0(1)	R0(2)	R9(1)	R9(2)	Result
H	H	L	X	Reset to 0000
H	H	X	L	Reset to 0000
X	X	H	H	Set to 1001 (9)
L	X	L	X	Counts
X	L	X	L	Counts

FUNCTION DIAGRAM — WHAT IS INSIDE



Ripple counter — the ÷2 and ÷5 sections are independent.
BCD decade: clock into CKA (14) and strap QA (12) to CKB (1); read QD QC QB QA.
VCC = pin 5, GND = pin 10 — not the usual corner pins.

PINOUT — DIP-16



DIP-16 — top view (notch / pin-1 dot at top)

AT A GLANCE

- Display type** COMMON ANODE only — outputs sink current, they do not source it
- Series resistors** Required on every segment (220–470 Ω at 5 V)
- LT (pin 3)** Lamp test: LOW turns every segment on
- RBI (pin 5)** Ripple-blank in: blanks a leading zero
- BI/RBO (pin 4)** Dual purpose: drive LOW to blank the digit, or read it as ripple-blank out
- Power** VCC = pin 16, GND = pin 8

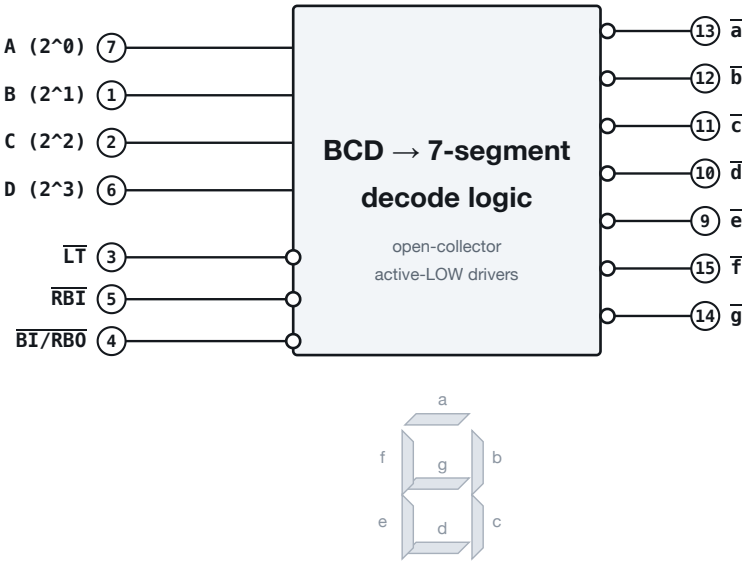
LOGIC TABLES

DECODED SEGMENTS (0 = SEGMENT ON / OUTPUT LOW)

Dec	D	C	B	A	a	b	c	d	e	f	g
0	0	0	0	0	0	0	0	0	0	0	1
1	0	0	0	1	1	0	0	1	1	1	1
2	0	0	1	0	0	0	1	0	0	1	0
3	0	0	1	1	0	0	0	0	1	1	0
4	0	1	0	0	1	0	0	1	1	0	0
5	0	1	0	1	0	1	0	0	1	0	0
6	0	1	1	0	1	1	0	0	0	0	0
7	0	1	1	1	0	0	0	1	1	1	1
8	1	0	0	0	0	0	0	0	0	0	0
9	1	0	0	1	0	0	0	1	1	0	0

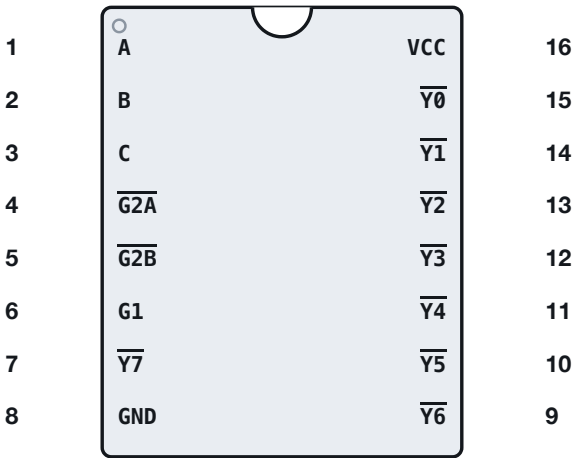
Inputs 10–15 produce unique but non-numeric patterns.

FUNCTION DIAGRAM — WHAT IS INSIDE



Drives a COMMON-ANODE display: each output sinks current when its segment is on.
Needs series resistors (typ. 220–470 Ω) per segment. VCC = 16, GND = 8.

PINOUT — DIP-16



DIP-16 — top view (notch / pin-1 dot at top)

AT A GLANCE

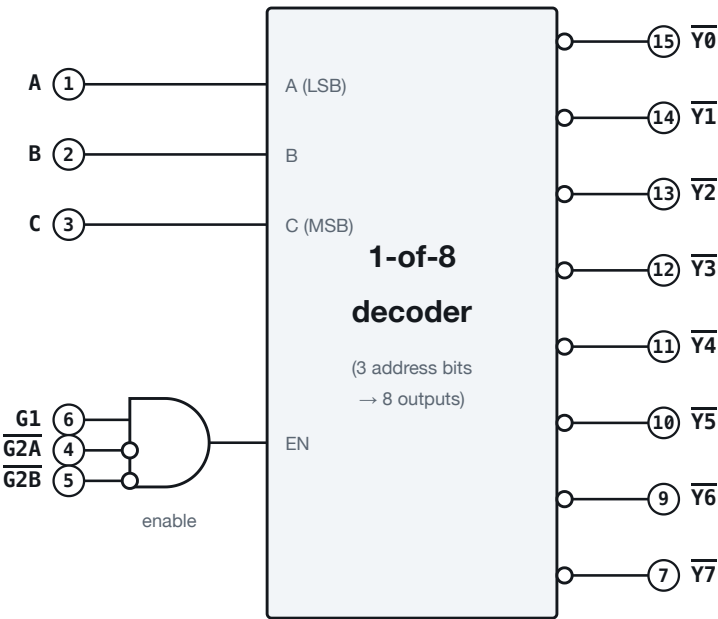
- Outputs** Active LOW — the selected line goes LOW, the other seven stay HIGH
- Address weight** A = LSB (pin 1), C = MSB (pin 3)
- Enables** G1 must be HIGH and both G2 inputs LOW
- Cascading** The three enables let you build a 1-of-24 or 1-of-32 decoder with no extra glue
- Power** VCC = pin 16, GND = pin 8

LOGIC TABLES

FUNCTION TABLE (H = HIGH, L = LOW, X = DON'T CARE)

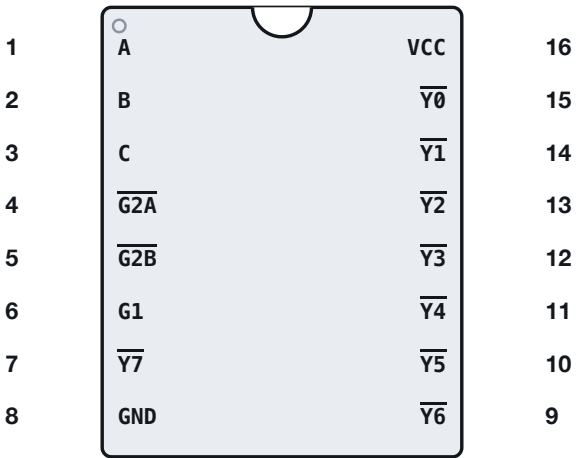
G1	G2A	G2B	C	B	A	Output LOW
L	X	X	X	X	X	none — all HIGH
X	H	X	X	X	X	none — all HIGH
X	X	H	X	X	X	none — all HIGH
H	L	L	0	0	0	Y0
H	L	L	0	0	1	Y1
H	L	L	0	1	0	Y2
H	L	L	0	1	1	Y3
H	L	L	1	0	0	Y4
H	L	L	1	0	1	Y5
H	L	L	1	1	0	Y6
H	L	L	1	1	1	Y7

FUNCTION DIAGRAM — WHAT IS INSIDE



Exactly one output goes LOW for each address; all others stay HIGH.
All outputs are HIGH unless G1 = 1 and G2A = G2B = 0. VCC = 16, GND = 8.

PINOUT — DIP-16



DIP-16 — top view (notch / pin-1 dot at top)

AT A GLANCE

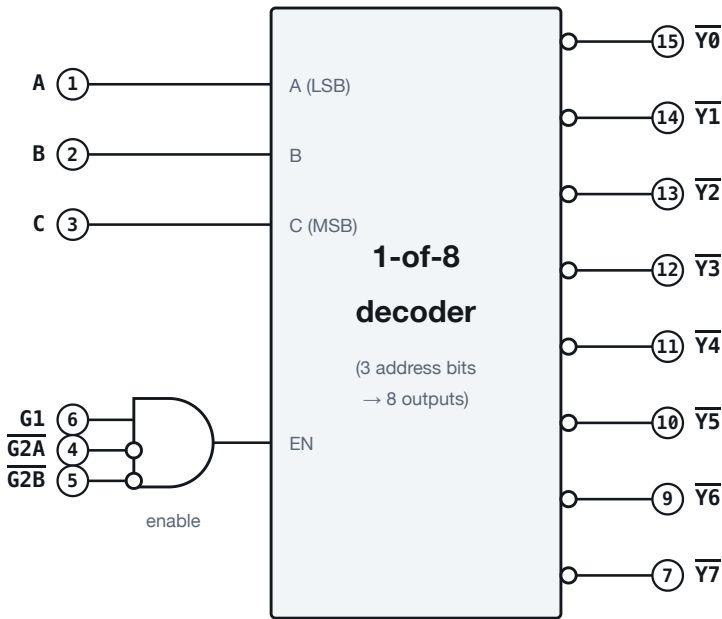
- Outputs** Active LOW — the selected line goes LOW, the other seven stay HIGH
- Address weight** A = LSB (pin 1), C = MSB (pin 3)
- Enables** G1 must be HIGH and both G2 inputs LOW
- Cascading** The three enables let you build a 1-of-24 or 1-of-32 decoder with no extra glue
- Power** VCC = pin 16, GND = pin 8

LOGIC TABLES

FUNCTION TABLE (H = HIGH, L = LOW, X = DON'T CARE)

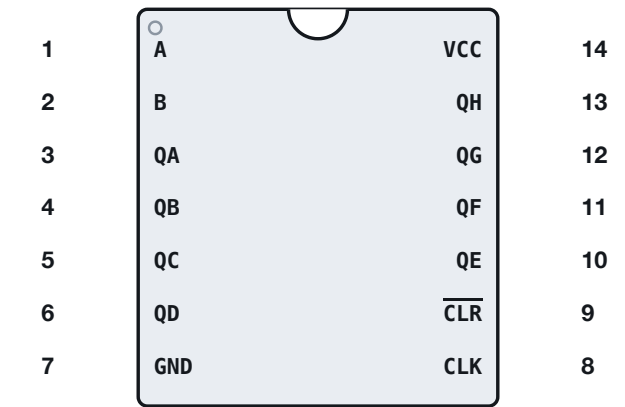
G1	G2A	G2B	C	B	A	Output LOW
L	X	X	X	X	X	none — all HIGH
X	H	X	X	X	X	none — all HIGH
X	X	H	X	X	X	none — all HIGH
H	L	L	0	0	0	Y0
H	L	L	0	0	1	Y1
H	L	L	0	1	0	Y2
H	L	L	0	1	1	Y3
H	L	L	1	0	0	Y4
H	L	L	1	0	1	Y5
H	L	L	1	1	0	Y6
H	L	L	1	1	1	Y7

FUNCTION DIAGRAM — WHAT IS INSIDE



Exactly one output goes LOW for each address; all others stay HIGH.
All outputs are HIGH unless G1 = 1 and G2A = G2B = 0. VCC = 16, GND = 8.

PINOUT — DIP-14

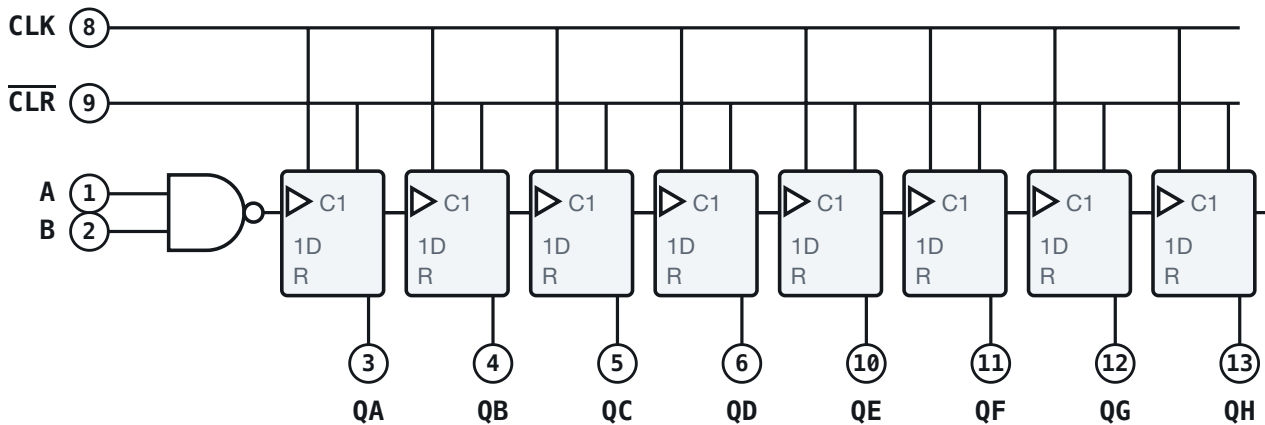


DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

- Serial input** A AND B — tie one HIGH and drive the other, or use B as a gate
- Trigger** Positive edge on CLK (pin 8)
- Clear** $\overline{\text{CLR}}$ (pin 9) LOW zeroes all eight stages, independent of the clock
- Typical use** Turn 2 microcontroller pins into 8 outputs (LED bars, keypad scan)
- Note** Outputs change while shifting — use a '595 instead if you need glitch-free outputs
- Power** VCC = pin 14, GND = pin 7

FUNCTION DIAGRAM — WHAT IS INSIDE



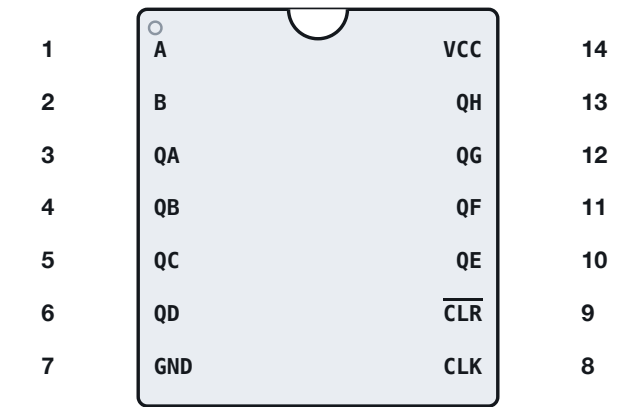
Serial-IN, parallel-OUT. Data at A-B is shifted one place on every rising CLK edge.
CLR = LOW clears all eight stages asynchronously. VCC = 14, GND = 7.

LOGIC TABLES

FUNCTION TABLE

$\overline{\text{CLR}}$	CLK	A	B	QA	QB ... QH
L	X	X	X	L	all L
H	L	X	X	hold	hold
H	↑	H	H	H	shifted right
H	↑	L	X	L	shifted right
H	↑	X	L	L	shifted right

PINOUT — DIP-14

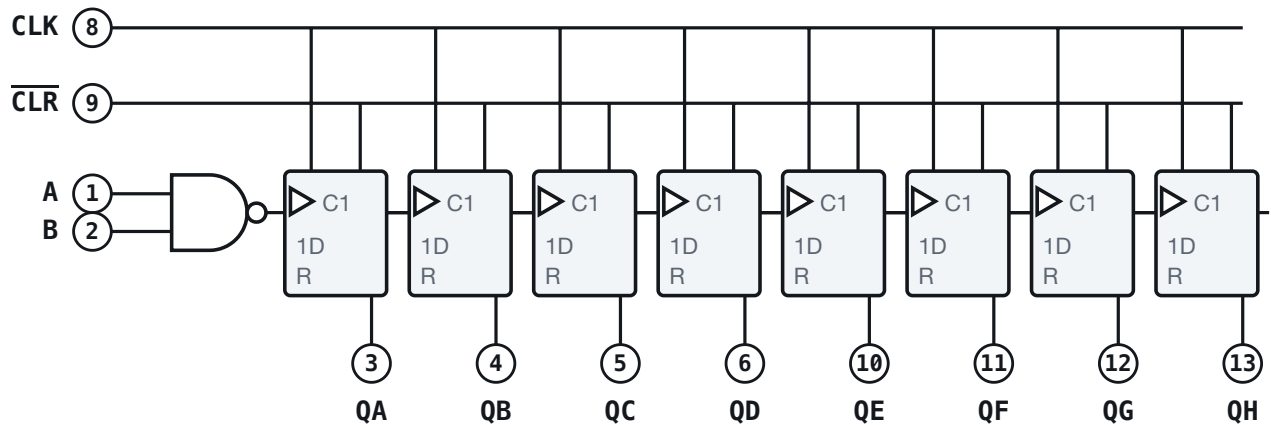


DIP-14 — top view (notch / pin-1 dot at top)

AT A GLANCE

- Serial input** A AND B — tie one HIGH and drive the other, or use B as a gate
- Trigger** Positive edge on CLK (pin 8)
- Clear** $\overline{\text{CLR}}$ (pin 9) LOW zeroes all eight stages, independent of the clock
- Typical use** Turn 2 microcontroller pins into 8 outputs (LED bars, keypad scan)
- Note** Outputs change while shifting — use a '595 instead if you need glitch-free outputs
- Power** VCC = pin 14, GND = pin 7

FUNCTION DIAGRAM — WHAT IS INSIDE



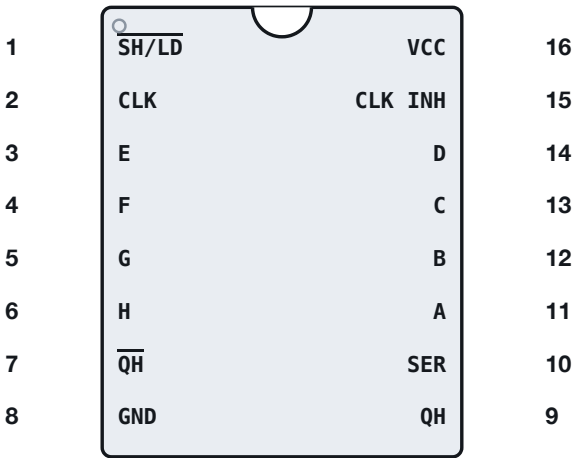
Serial-IN, parallel-OUT. Data at A-B is shifted one place on every rising CLK edge.
CLR = LOW clears all eight stages asynchronously. VCC = 14, GND = 7.

LOGIC TABLES

FUNCTION TABLE

$\overline{\text{CLR}}$	CLK	A	B	QA	QB ... QH
L	X	X	X	L	all L
H	L	X	X	hold	hold
H	↑	H	H	H	shifted right
H	↑	L	X	L	shifted right
H	↑	X	L	L	shifted right

PINOUT — DIP-16



DIP-16 — top view (notch / pin-1 dot at top)

AT A GLANCE

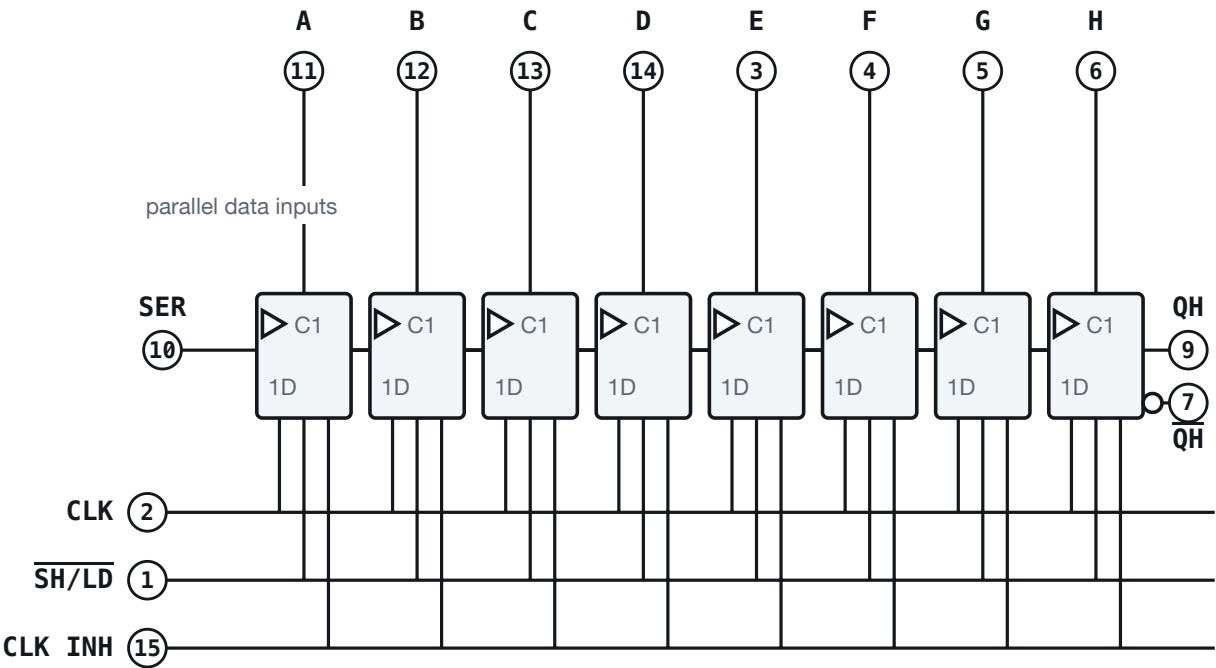
- Direction** A is loaded into the first stage, H into the last; H appears at QH first
- Serial input** SER (pin 10) — feed the previous chip's QH here to cascade
- Outputs** QH (pin 9) and its complement $\overline{QH}$ (pin 7)
- Typical use** Read 8 buttons/switches with 3 microcontroller pins
- Reading order** Pulse $\overline{SH/LD}$ LOW then HIGH to snapshot the inputs, then clock 8 times
- Power** VCC = pin 16, GND = pin 8

LOGIC TABLES

FUNCTION TABLE

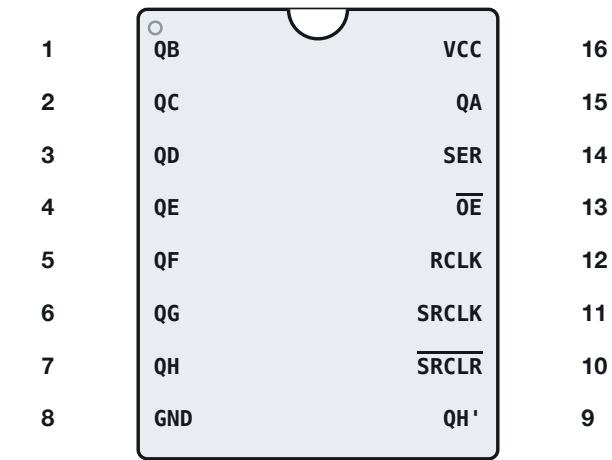
SH/LD	CLK INH	CLK	Action
L	X	X	Parallel load A–H (asynchronous)
H	L	↑	Shift one place toward QH
H	L	L	Hold
H	H	X	Hold – clock inhibited

FUNCTION DIAGRAM — WHAT IS INSIDE



Parallel-IN, serial-OUT. $\overline{SH/LD}$ LOW loads A–H asynchronously; HIGH shifts on each rising CLK. CLK INH HIGH freezes the register. H reaches QH first; cascade QH into the next chip's SER. VCC = pin 16, GND = pin 8.

PINOUT — DIP-16



DIP-16 — top view (notch / pin-1 dot at top)

AT A GLANCE

- Why two clocks
- SRCLK shifts, RCLK publishes — outputs stay stable while data is shifting in
- Cascading
- QH' (pin 9) is the overflow bit; chain it to the next chip's SER
- Drive
- ±6 mA per output — fine for an LED with a series resistor
- Power
- VCC = pin 16, GND = pin 8

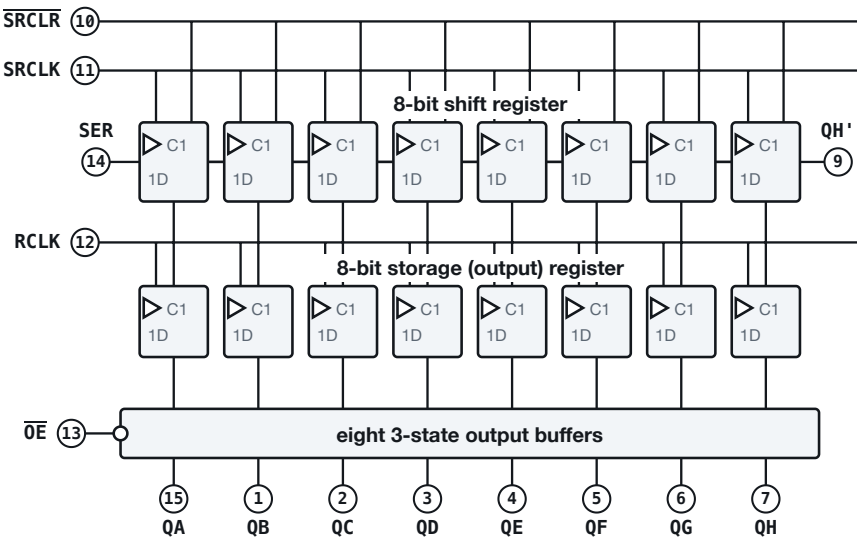
LOGIC TABLES

FUNCTION TABLE				
SRCLR	SRCLK	RCLK	OE	Action
L	X	X	X	Clear the shift register
H	↑	L	L	Shift SER in one place
H	L	↑	L	Copy shift register → output latch
H	X	X	H	QA–QH go high-impedance (latch keeps its data)

TYPICAL WIRING TO A MICROCONTROLLER

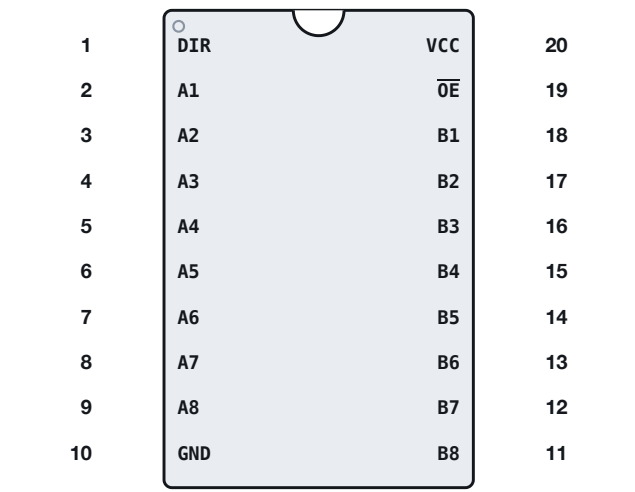
Pin	Name	Connect to
14	SER	Data out
11	SRCLK	Clock
12	RCLK	Latch / strobe
10	SRCLR	VCC (tie HIGH if unused)
13	OE	GND (tie LOW if unused)
9	QH'	SER of the next '595

FUNCTION DIAGRAM — WHAT IS INSIDE



SER shifts in on each rising SRCLK. A rising RCLK copies the shift register into the storage register, so the outputs never glitch while data is being shifted in. OE HIGH → QA–QH high-Z. QH' is the overflow bit — chain it to the next '595 SER. VCC = pin 16, GND = pin 8.

PINOUT — DIP-20



DIP-20 — top view (notch / pin-1 dot at top)

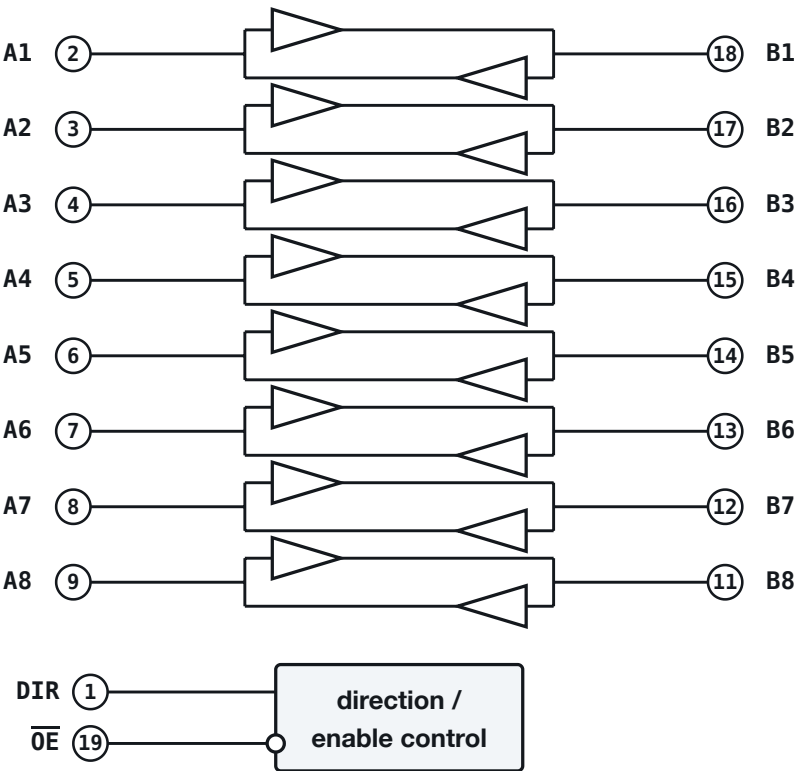
AT A GLANCE

- Channels**8 bidirectional, non-inverting
- Drive**Up to 24 mA sink — designed to drive a loaded backplane
- Careful**Never enable two drivers onto the same bus at once — bus contention
- Typical use**Level buffering / direction switching between a CPU and a peripheral bus
- Power**VCC = pin 20, GND = pin 10

LOGIC TABLES

FUNCTION TABLE		
$\overline{OE}$	DIR	Operation
L	H	A data → B bus
L	L	B data → A bus
H	X	Both buses isolated (high-Z)

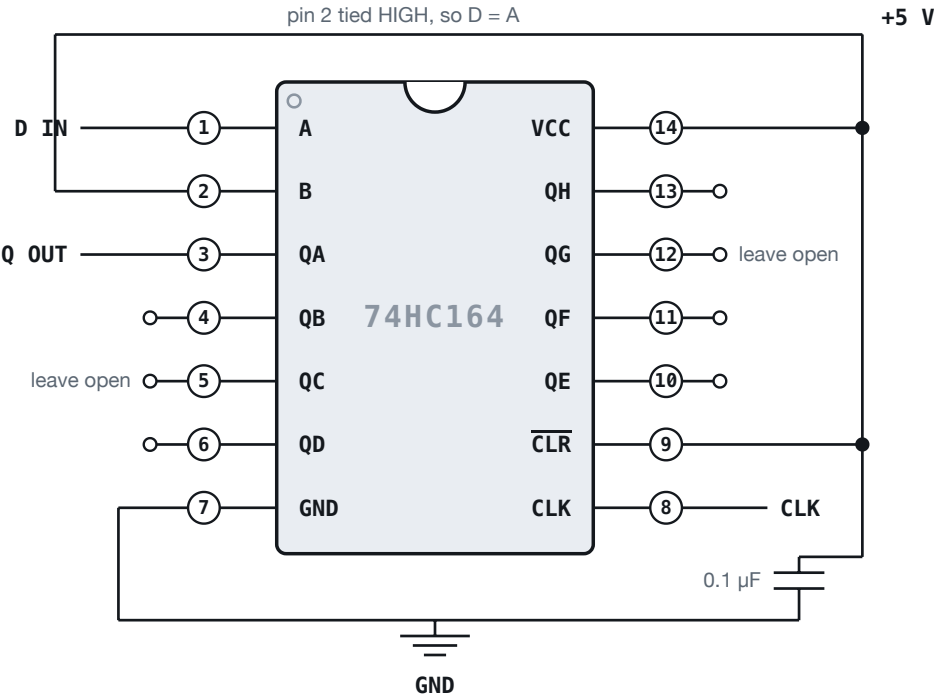
FUNCTION DIAGRAM — WHAT IS INSIDE



$\overline{OE}$ = HIGH → both A and B buses are high-impedance.
 $\overline{OE}$ = LOW and DIR = HIGH → A drives B. $\overline{OE}$ = LOW and DIR = LOW → B drives A.
VCC = pin 20, GND = pin 10.

This kit has no 74HC74, so the only dedicated D flip-flop is the bipolar 74LS74 on page 15. When you want a CMOS flip-flop — and want to keep the '74's preset and clear free for set/reset work later — take the **first stage of a 74HC164**. Each of its eight stages already *is* a positive-edge D flip-flop; you simply use stage A and ignore the rest — its full internal view is on page 21.

WIRING — DIP-14, TOP VIEW



CONNECTIONS

Pin	Name	Wire it to
1	A	D input
2	B	VCC — makes D = A
3	QA	Q output
8	CLK	Clock, rising edge
9	CLR	VCC — LOW clears Q
14	VCC	+5 V (2 – 6 V)
7	GND	0 V
4–6, 10–13	QB...QH	Leave open

RESULTING FLIP-FLOP

CLR	CLK	D	Q
L	X	X	L
H	↑	H	H
H	↑	L	L
H	L	X	hold

Positive-edge triggered, exactly like a '74 — but with no /Q output.

WORTH KNOWING

- No /Q

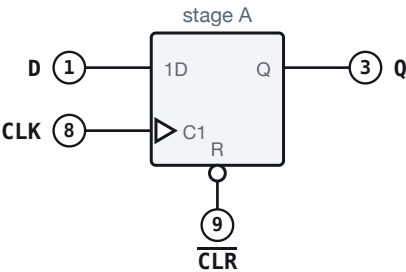
The '164 brings out only true outputs. If you need the complement, invert QA through a spare gate on the 74HC04 or 74HC00.
- Free async reset

/CLR on pin 9 is already an active-LOW asynchronous clear. Tie it HIGH to ignore it, or drive it and you have D, CLK, Q *and* reset.
- Free delay taps

QB...QH are the same data 2...8 clocks later — useful as an edge synchroniser or a shift-register debounce without adding a part.
- Why not the others

The 74HC595's first stage is not brought out — its serial output is 8 clocks late — and the 74HC165 loads the wrong direction.

WHAT YOU ARE LEFT WITH



Numbers in circles are 74HC164 package pins.